

Week 5_Practical (Name-Kamtam Priyanka, 2520090001)

Task: Library Backend & Security API using FastAPI, MySQL, JWT Authentication and Async SQLAlchemy

Project Structure:

Week_5_library_fastapi_project

- |— main.py
- |— database.py
- |— models.py
- |— schemas.py
- |— security.py
- |— embedding.py
- |— requirements.txt

MySQL Workbench Code:

```
CREATE DATABASE library_db;
```

```
USE library_db;
```

```
CREATE TABLE IF NOT EXISTS members (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    username VARCHAR(100) UNIQUE NOT NULL,  
    password_hash VARCHAR(255) NOT NULL,  
    role VARCHAR(30) NOT NULL DEFAULT 'member'  
);
```

```
CREATE TABLE IF NOT EXISTS books (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    title VARCHAR(250) NOT NULL,  
    author VARCHAR(200) NOT NULL,  
    isbn VARCHAR(13) UNIQUE NOT NULL,
```

```
    price DECIMAL(10,2) NOT NULL,  
    published_date DATE NOT NULL,  
    description TEXT,  
    embedding JSON  
);
```

database.py

```
from sqlalchemy.ext.asyncio import (  
    AsyncSession, async_sessionmaker, create_async_engine  
)  
from sqlalchemy.orm import DeclarativeBase  
DATABASE_URL = (  
    "mysql+asyncmy://root:YOUR_MYSQL_PASSWORD"  
    "@localhost:3306/library_db"  
)  
engine = create_async_engine(DATABASE_URL, echo=True)  
AsyncSessionLocal = async_sessionmaker(  
    engine, expire_on_commit=False  
)  
class Base(DeclarativeBase):  
    pass  
async def get_db():  
    async with AsyncSessionLocal() as session:  
        yield session
```

models.py

```
from datetime import date  
from decimal import Decimal
```

```
from sqlalchemy import Date, Integer, Numeric, String, Text, JSON
```

```
from sqlalchemy.orm import Mapped, mapped_column
```

```
from database import Base
```

```
class Member(Base):
```

```
    __tablename__ = "members"
```

```
    id: Mapped[int] = mapped_column(
```

```
        Integer, primary_key=True
```

```
)
```

```
    username: Mapped[str] = mapped_column(
```

```
        String(100), unique=True, nullable=False
```

```
)
```

```
    password_hash: Mapped[str] = mapped_column(
```

```
        String(255), nullable=False
```

```
)
```

```
    role: Mapped[str] = mapped_column(
```

```
        String(30), default="member", nullable=False
```

```
)
```

```
class Book(Base):
```

```
    __tablename__ = "books"
```

```
    id: Mapped[int] = mapped_column(
```

```
        Integer, primary_key=True
```

```
)
```

```
    title: Mapped[str] = mapped_column(
```

```
        String(250), nullable=False
```

```
)
```

```
    author: Mapped[str] = mapped_column(
```

```

        String(200), nullable=False
    )
    isbn: Mapped[str] = mapped_column(
        String(13), unique=True, nullable=False
    )
    price: Mapped[Decimal] = mapped_column(
        Numeric(10, 2), nullable=False
    )
    published_date: Mapped[date] = mapped_column(
        Date, nullable=False
    )
    description: Mapped[str | None] = mapped_column(
        Text, nullable=True
    )
    embedding: Mapped[list[float] | None] = mapped_column(
        JSON, nullable=True
    )

```

schemas.py

```

from datetime import date
from decimal import Decimal
from pydantic import BaseModel, Field, field_validator
class BookSchema(BaseModel):
    title: str
    author: str
    isbn: str = Field(
        min_length=13,

```

```

        max_length=13
    )
    price: Decimal = Field(gt=0)
    published_date: date
    description: str | None = None
    @field_validator("published_date")
    @classmethod
    def validate_published_date(cls, value: date):
        if value > date.today():
            raise ValueError(
                "published_date cannot be in the future"
            )
        return value
class MemberRegister(BaseModel):
    username: str
    password: str = Field(min_length=6)
class TokenResponse(BaseModel):
    access_token: str
    token_type: str

```

security.py

```

from datetime import datetime, timedelta, timezone
from fastapi import Depends, HTTPException
from fastapi.security import OAuth2PasswordBearer
from jose import JWTError, jwt
from passlib.context import CryptContext
from sqlalchemy import select

```

```
from sqlalchemy.ext.asyncio import AsyncSession
from database import get_db
from models import Member

SECRET_KEY = "CHANGE_THIS_SECRET"
ALGORITHM = "HS256"
ACCESS_TOKEN_EXPIRE_MINUTES = 60

pwd_context = CryptContext(
    schemes=["bcrypt"],
    deprecated="auto"
)

oauth2_scheme = OAuth2PasswordBearer(
    tokenUrl="/login"
)

def hash_password(password: str):
    return pwd_context.hash(password)

def verify_password(
    plain_password,
    hashed_password
):
    return pwd_context.verify(
        plain_password,
        hashed_password
    )

def create_access_token(username, role):
    expire = datetime.now(timezone.utc) + timedelta(
        minutes=ACCESS_TOKEN_EXPIRE_MINUTES
```

```

    )
    payload = {
        "sub": username,
        "role": role,
        "exp": expire
    }
    return jwt.encode(
        payload,
        SECRET_KEY,
        algorithm=ALGORITHM
    )

async def get_current_member(
    token: str = Depends(oauth2_scheme),
    db: AsyncSession = Depends(get_db)
):
    credentials_exception = HTTPException(
        status_code=401,
        detail="Invalid or expired token",
        headers={
            "WWW-Authenticate": "Bearer"
        }
    )
    try:
        payload = jwt.decode(
            token,
            SECRET_KEY,

```

```

        algorithms=[ALGORITHM]
    )
    username = payload.get("sub")
    if not username:
        raise credentials_exception
except JWTError:
    raise credentials_exception
result = await db.execute(
    select(Member).where(
        Member.username == username
    )
)
member = result.scalar_one_or_none()
if not member:
    raise credentials_exception
return member

```

embedding.py

```

import hashlib
import math
VECTOR_DIM = 384
def text_to_vector(text: str) -> list[float]:
    values = []
    for i in range(VECTOR_DIM):
        digest = hashlib.sha256(
            f"{text.lower()}:{i}".encode()
        ).digest()

```

```

        number = int.from_bytes(
            digest[:4],
            byteorder="big"
        )
        values.append(
            (number / 2**32) * 2 - 1
        )
    magnitude = math.sqrt(
        sum(x * x for x in values)
    )
    return [
        x / magnitude
        for x in values
    ]

```

main.py

```

from fastapi import (
    BackgroundTasks,
    Depends,
    FastAPI,
    HTTPException,
    Query,
    status
)
from fastapi.security import OAuth2PasswordRequestForm
from sqlalchemy import select
from sqlalchemy.ext.asyncio import AsyncSession
from database import Base, engine, get_db
from models import Book, Member
from schemas import BookSchema, MemberRegister, TokenResponse
from security import (
    create_access_token,
    get_current_member,

```

```

    hash_password,
    verify_password
)
from embedding import text_to_vector
import math
app = FastAPI(
    title="Library Backend & Security API",
    version="1.0"
)
# Create tables when application starts
@app.on_event("startup")
async def startup():
    async with engine.begin() as connection:
        await connection.run_sync(
            Base.metadata.create_all
        )
# Home
@app.get("/")
async def home():
    return {
        "message": "Library API is running"
    }
# Background task for logging
def log_new_book(book_id: int, title: str):
    with open(
        "book_events.txt",
        "a",
        encoding="utf-8"
    ) as file:
        file.write(
            f"New Book Added: id={book_id}, "
            f"title={title}\n"
        )
# Cosine similarity
def cosine_similarity(
    a: list[float],
    b: list[float]
) -> float:
    if len(a) != len(b):
        return -1.0

```

```

dot = sum(
    x * y
    for x, y in zip(a, b)
)
norm_a = math.sqrt(
    sum(x * x for x in a)
)
norm_b = math.sqrt(
    sum(y * y for y in b)
)
if norm_a == 0 or norm_b == 0:
    return 0.0
return dot / (norm_a * norm_b)
# Register member
@app.post("/register")
async def register(
    member: MemberRegister,
    db: AsyncSession = Depends(get_db)
):
    result = await db.execute(
        select(Member).where(
            Member.username == member.username
        )
    )
    if result.scalar_one_or_none():
        raise HTTPException(
            status_code=400,
            detail="Username already exists"
        )
    new_member = Member(
        username=member.username,
        password_hash=hash_password(
            member.password
        ),
        role="member"
    )
    db.add(new_member)
    await db.commit()
    await db.refresh(new_member)
    return {

```

```

        "message": "Member registered successfully",
        "username": new_member.username,
        "role": new_member.role
    }
# Login
@app.post(
    "/login",
    response_model=TokenResponse
)
async def login(
    form_data: OAuth2PasswordRequestForm = Depends(),
    db: AsyncSession = Depends(get_db)
):
    result = await db.execute(
        select(Member).where(
            Member.username == form_data.username
        )
    )
    member = result.scalar_one_or_none()
    if (
        not member
        or not verify_password(
            form_data.password,
            member.password_hash
        )
    ):
        raise HTTPException(
            status_code=401,
            detail="Invalid username or password"
        )
    token = create_access_token(
        member.username,
        member.role
    )
    return {
        "access_token": token,
        "token_type": "bearer"
    }
# Add book - protected by JWT
@app.post(

```

```

    "/books",
    status_code=status.HTTP_201_CREATED
)
async def create_book(
    book: BookSchema,
    background_tasks: BackgroundTasks,
    current_member: Member = Depends(
        get_current_member
    ),

    db: AsyncSession = Depends(get_db)
):
    # Create embedding
    embedding = text_to_vector(
        f"{book.title} "
        f"{book.author} "
        f"{book.description or ""}"
    )
    new_book = Book(
        title=book.title,
        author=book.author,
        isbn=book.isbn,
        price=book.price,
        published_date=book.published_date,
        description=book.description,
        embedding=embedding
    )
    db.add(new_book)
    await db.commit()
    await db.refresh(new_book)
    # Schedule background logging
    background_tasks.add_task(
        log_new_book,
        new_book.id,
        new_book.title
    )
    return {
        "message": "Book added successfully",
        "book_id": new_book.id,
        "added_by": current_member.username

```

```

    }
# Search books using cosine similarity
@app.get("/books/search")
async def search_books(
    q: str = Query(min_length=1),
    limit: int = Query(
        default=5,
        ge=1,
        le=20
    ),
    current_member: Member = Depends(
        get_current_member
    ),
    db: AsyncSession = Depends(get_db)
):
    query_vector = text_to_vector(q)
    result = await db.execute(
        select(Book).where(
            Book.embedding.is_not(None)
        )
    )

    books = result.scalars().all()
    ranked = []
    for book in books:
        if book.embedding:

            similarity = cosine_similarity(
                query_vector,
                book.embedding
            )
            ranked.append({
                "id": book.id,
                "title": book.title,
                "author": book.author,
                "isbn": book.isbn,
                "price": float(book.price),
                "published_date":
                    book.published_date,
                "description":

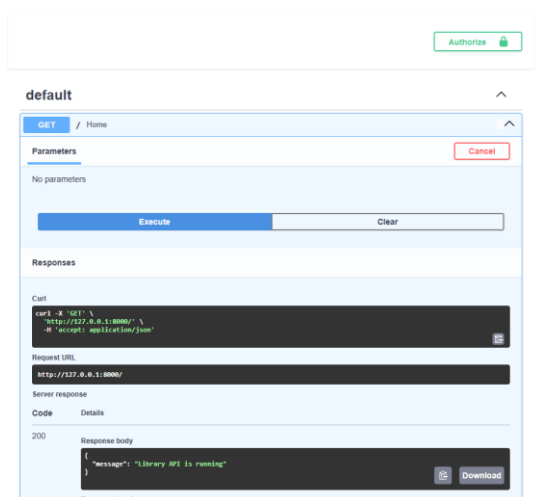
```

```

        book.description,
        "similarity":
            similarity
    })
    ranked.sort(
        key=lambda item: item["similarity"],
        reverse=True
    )
    return {
        "query": q,
        "results": ranked[:limit]
    }
}

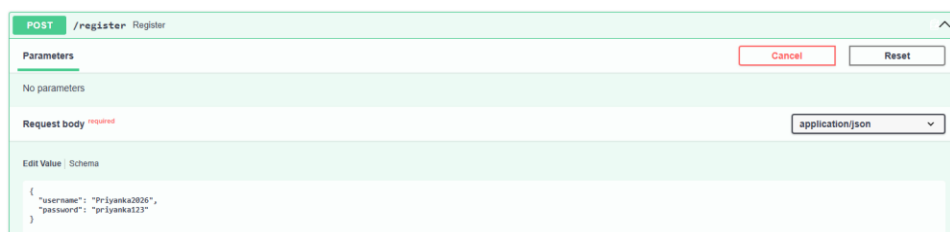
```

GET:



1. Register

POST /register



After execution:

Responses

Curl

```
curl -X 'POST' \
  http://127.0.0.1:8080/register \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "Priyanka2026",
    "password": "priyanka123"
  }'
```

Request URL

http://127.0.0.1:8080/register

Server response

Code Details

200

Response body

```
{
  "message": "Number registered successfully",
  "username": "Priyanka2026",
  "role": "member"
}
```

Response headers

```
content-length: 86
content-type: application/json
date: Sat, 22 Sep 2024 05:26:15 GMT
server: uvicorn
```

Authorization:

Available authorizations x

Scopes are used to grant an application different levels of access to data on behalf of the end user. Each API may declare one or more scopes.
API requires the following scopes. Select which ones you want to grant to Swagger UI.

OAuth2PasswordBearer (OAuth2, password)

Authorized

Token URL: /login
Flow: password
username: Priyanka2026
password: *****
Client credentials location: basic
client_secret: *****

[Logout](#) [Close](#)

2. Login

POST /login

POST /login Login

Parameters

No parameters

Request body required

application/x-www-form-urlencoded

grant_type

string | (string | null)

pattern: "password"

☐ Send empty value

username required

string

password required

string(\$password)

scope

string

☒ Send empty value

client_id

string | (string | null)

☐ Send empty value

client_secret

string | (string | null)(\$password)

☐ Send empty value

Response:

Responses

Curl

```
curl -X 'POST' \  
  'http://127.0.0.1:8000/login' \  
  -H 'accept: application/json' \  
  -H 'Content-Type: application/x-www-form-urlencoded' \  
  -d 'grant_type=password&username=Priyanka2026&password=priyanka123&scope=&client_id=string&client_secret=*****'
```

Request URL

```
http://127.0.0.1:8000/login
```

Server response

CodeDetails

200

Response body

```
{  
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IWRkcS9eYjZkdTI0IiwiaWF0IjEybmVtYWwYlWwiIiwiaWF0IjEwNzg5MTkxMjYyLWVybXNr4QcLbnguUlPa4T_g4ydi75bCJBoal3jmu0",  
  "token_type": "bearer"  
}
```

Download

Response headers

```
content-length: 196  
content-type: application/json  
date: Sat, 12 Sep 2026 05:30:51 GMT  
server: unicorn
```

3. POST /books

POST

/books Create Book

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value

Schema

```
{
  "title": "Priyanka's Guide to Java",
  "author": "Priyanka",
  "isbn": "9781345678901",
  "price": 599.00,
  "published_date": "2025-08-15",
  "description": "A beginner friendly guide to Java programming and data structures."
}
```

Response:

The screenshot shows a REST client interface with a GET request to `http://127.0.0.1:8080/books`. The response is a JSON array containing one book object. The book object has the following properties: `id` (1), `author` ("Prityamk"), `title` ("Prityamk's Guide to Java"), `price` (599.99), `available` (true), and `description` ("A beginner friendly guide to Java programming and data structures.")

```
curl -X GET 'http://127.0.0.1:8080/books'
{
  "id": 1,
  "author": "Prityamk",
  "title": "Prityamk's Guide to Java",
  "price": 599.99,
  "available": true,
  "description": "A beginner friendly guide to Java programming and data structures."
}
```

Request URL: `http://127.0.0.1:8080/books`

Server response

Code: 201 Details

Response body

```
{
  "message": "Book added successfully",
  "book_id": 1,
  "added_by": "Prityamk@KDS"
}
```

Response headers

```
content-length: 75
content-type: application/json
date: Sat, 12 Sep 2020 05:12:53 GMT
server: mule4
```

4. GET /books/search

GET

/books/search

Search Books

Parameters

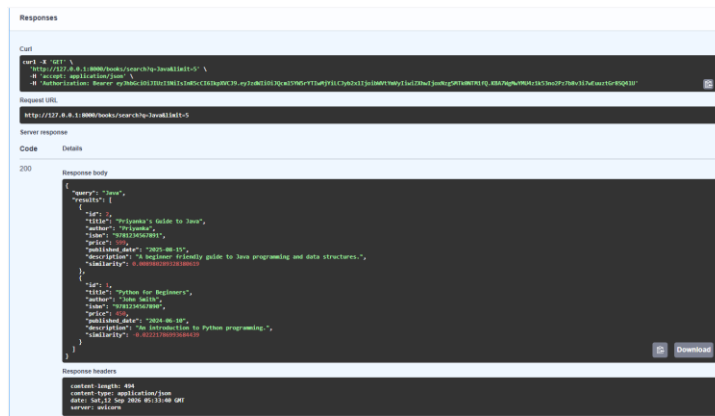
Cancel

Name	Description
q * required string (query) Java minLength: 1	
limit integer (query) 5 max: 200 min: 0	

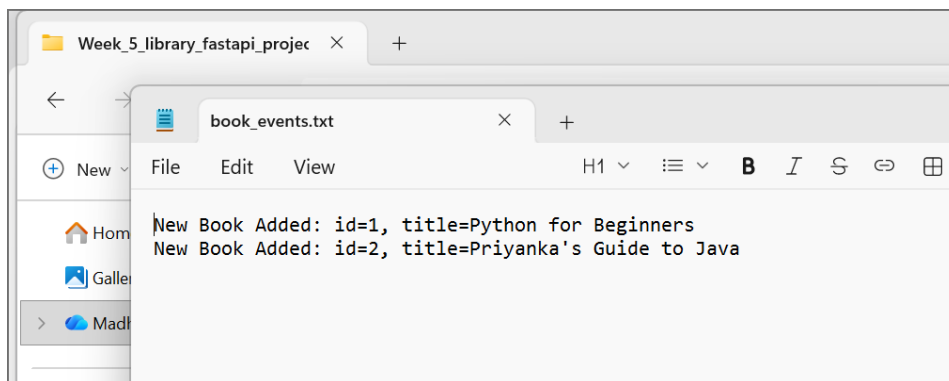
Execute

Clear

Response:

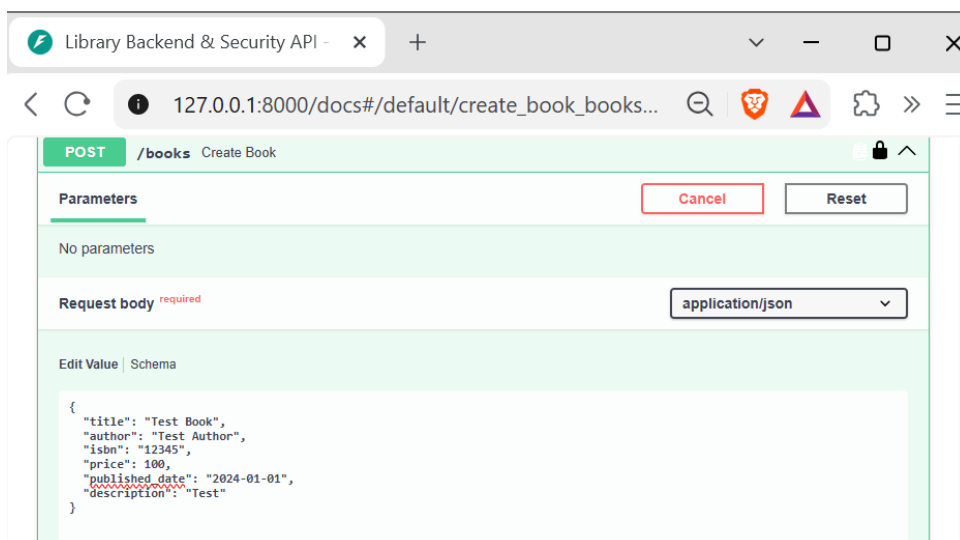


6. Check the background task



7. Test validation

Test A — Invalid ISBN



Response will be 422 error:

Server response

Code	Details
422	Error: Unprocessable Entity Response body <pre>{ "detail": [{ "type": "string_too_short", "loc": ["body", "isbn"], "msg": "String should have at least 13 characters", "input": "12345", "ctx": { "min_length": 13 } }]}</pre>

Download

Test B — Negative price

POST /books Create Book

Parameters Cancel Reset

No parameters

Request body required application/json

Edit Value Schema

```
{  "title": "Test Book",  "author": "Test Author",  "isbn": "1234567890100",  "price": -100,  "published_date": "2024-01-01",  "description": "Test"}
```

Response will be 422 since $\text{price} < 0$:

Code	Details
422	Error: Unprocessable Entity Response body <pre>{ "detail": [{ "type": "greater_than", "loc": ["body", "price"], "msg": "Input should be greater than 0", "input": -100, "ctx": { "gt": 0 } }]}</pre>

Download

Test C — Future publication date

POST /books Create Book

Parameters Cancel Reset

No parameters

Request body required application/json

Edit Value Schema

```
{  "title": "Future Book",  "author": "Test Author",  "isbn": "1234567890124",  "price": 100,  "published_date": "2035-01-01",  "description": "Test"}
```

Response will be 422 error because a book cannot have a future publication date:

Server response	
Code	Details
422	Error: Unprocessable Entity Response body <pre>{ "detail": [{ "type": "value_error", "loc": ["body", "published_date"], "msg": "Value error, published_date cannot be in the future", "input": "2035-01-01", "ctx": { "error": {} } }] }</pre>  Download